

Versi Dokumen: 1.0.0

Tanggal Pembuatan: 2026-06-29

Source Basis Repo: `firmware/platformio.ini`, `firmware/shared/include/*.h`, `firmware/config/*.h`, `firmware/gld/`, `firmware/ch/`, `firmware/gateway/`, `docs/design/*/final_design.md`, `server/nodered/`

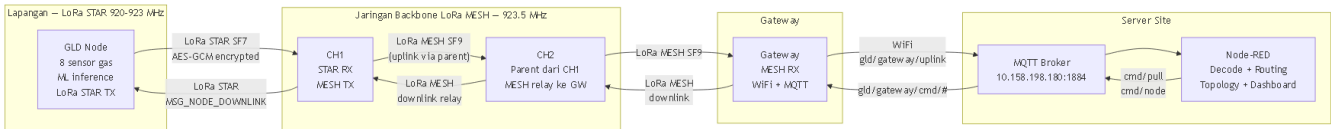
Status Firmware: GLD 0.8.12 · CH 0.7.1 · Gateway 0.1.3 · Protocol 0.1.0

Daftar Isi

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1. Visual Executive Summary

Alur Data GLD ke Server (dan Downlink Server ke GLD)



Gambar 1: Alur data GLD ke Server (uplink) dan Server ke GLD (downlink). CH tersusun parent-child — semua traffic MESH melewati jalur parent yang sama.

Tabel Ringkasan Komponen

Komponen	Perangkat	Firmware	Peran Utama
GLD	ESP32-WROOM-1U	0.8.12	Sensor gas, ML inference, enkripsi, LoRa STAR TX

Komponen	Perangkat	Firmware	Peran Utama
CH1	ESP32-WROOM-1U	0.7.1	STAR RX (dari GLD), NodeCache, MESH TX ke parent
CH2	ESP32-WROOM-1U	0.7.1	Parent dari CH1, MESH relay ke Gateway
Gateway	ESP32-WROOM-1U	0.1.3	MESH RX, MQTT bridge ke Server Site
Server Dataset	PC/Server	—	MQTT Broker Dataset + Node-RED untuk capture data training
Server Site	PC/Server	—	MQTT Broker Site + Node-RED untuk monitoring, alarm, topologi

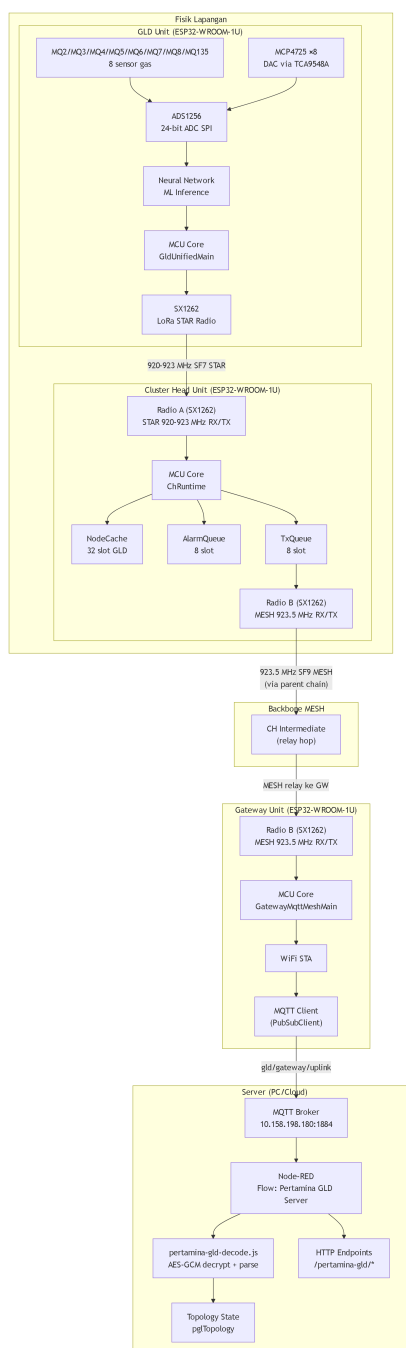
Ringkasan Alur Data Utama

Alur	Trigger	Jalur Lengkap
Normal Cache Update	Tiap 10 detik	GLD ke CH (simpan di NodeCache, tidak diteruskan otomatis ke server)
Server Pull Data	Node-RED request	Node-RED ke MQTT ke GW ke CH (hopList relay) ke GW ke MQTT ke Node-RED
Alarm Push	GasClass tidak sama dengan 0 dan conf lebih dari sama dengan 40	GLD ke CH ke parent chain CH ke GW ke MQTT ke Node-RED (langsung, tanpa pull request)
Dataset Capture	Menerima START_DATASET via MQTT	GLD ke MQTT Broker Dataset ke Node-RED
Downlink Command	Operator	Node-RED ke MQTT ke GW ke CH ke GLD (via LoRa STAR)
Topology Report	CH boot + tiap 5 menit	CH ke parent CHx ke GW ke MQTT ke Node-RED
Nulling	Mode nulling, offline	GLD lokal saja (DAC/ADS), tidak ke server

■ **Penting:** Data sensor GLD **non-alarm** tidak otomatis diteruskan ke server — CH hanya menyimpan di NodeCache dan baru dikirim ke server saat ada pull request dari Node-RED. Pengecualian: **data alarm** langsung di-push dari CH ke Gateway ke Node-RED tanpa perlu pull request.

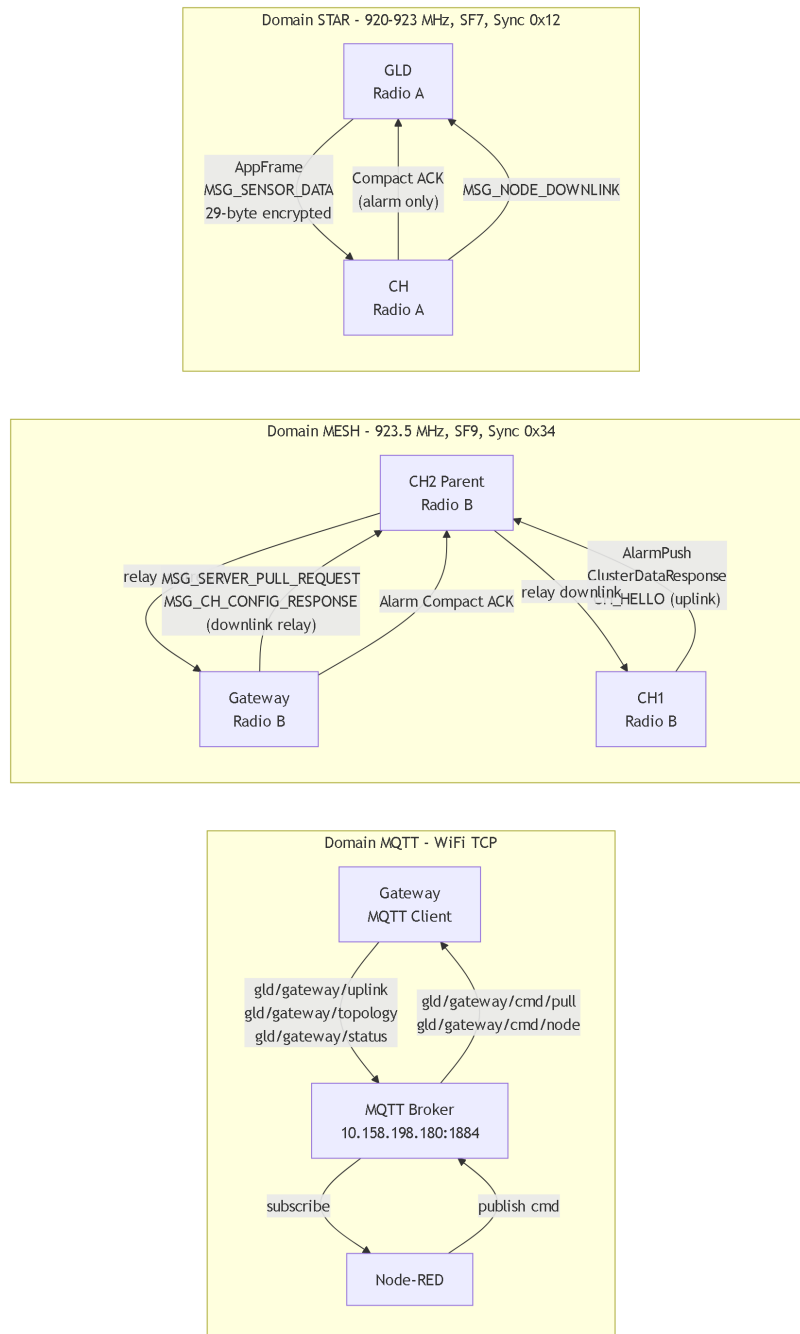
2. Peta Arsitektur Sistem

Diagram Arsitektur End-to-End



Gambar 2: Arsitektur end-to-end lengkap dengan komponen internal setiap node.

Diagram Jalur Komunikasi



Gambar 3: Tiga domain komunikasi yang digunakan sistem Pertamina GLD.

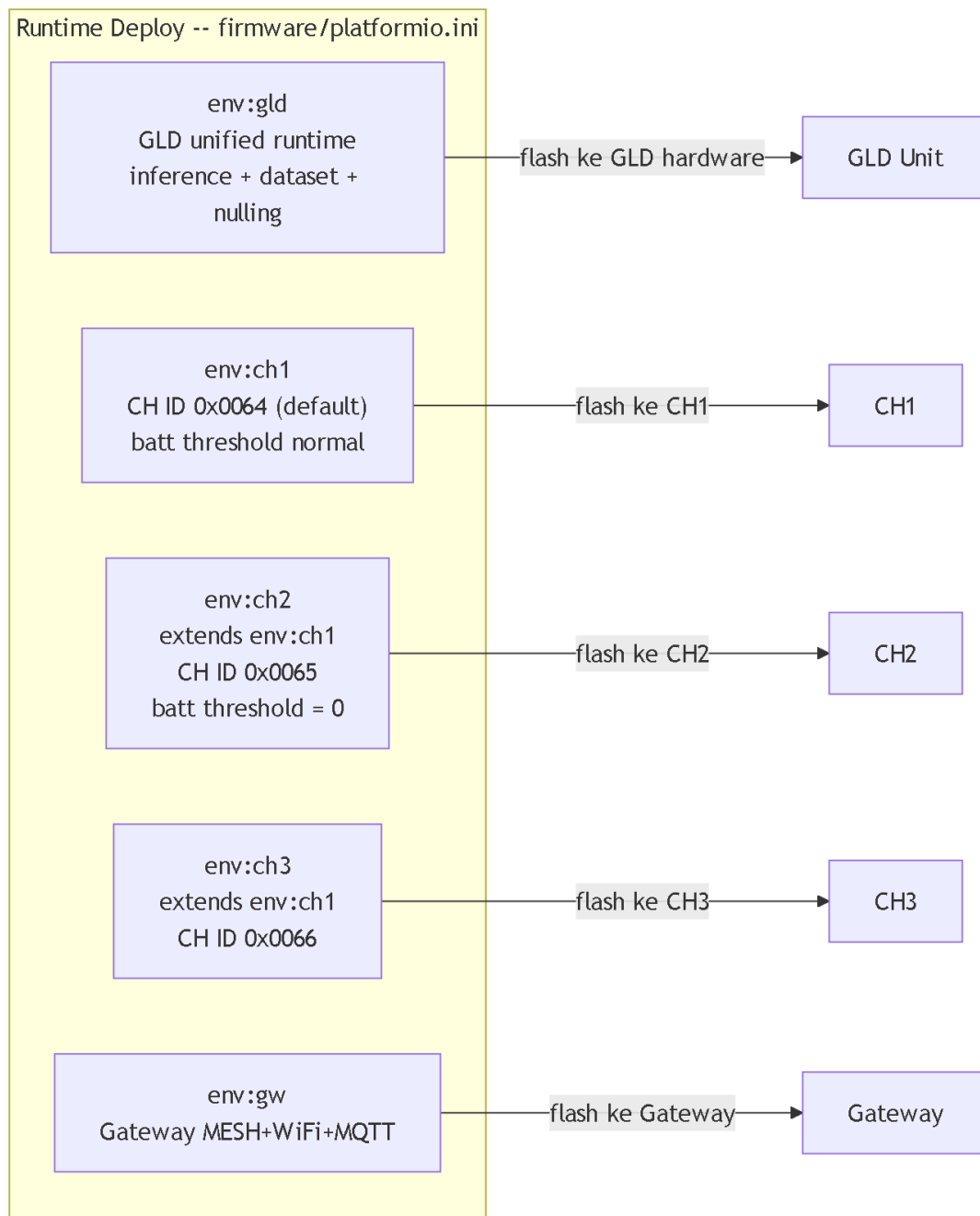
Penjelasan Per Jalur

Jalur	Frekuensi	SF	Sync Word	Max Payload	Arah
LoRa STAR	920-923 MHz (per-CH berbeda)	SF7	0x12	64 bytes	GLD ke CH (uplink), CH ke GLD (downlink)
LoRa MESH	923.5 MHz	SF9	0x34	80 bytes	CH ke CH ke Gateway (bidirectional)
MQTT/LAN	WiFi TCP	—	—	1024 bytes	Gateway ke Server

■ **Catatan:** Frekuensi deployment di atas adalah nilai operasional. Source config saat ini menggunakan 920.0 MHz (STAR) dan 921.0 MHz (MESH) — perlu diupdate sesuai deployment. Dua domain LoRa menggunakan sync word berbeda agar tidak saling mengganggu. CH memiliki dua radio fisik terpisah (Radio A = STAR, Radio B = MESH).

3. Peta Firmware Environment

Visual Map PlatformIO



Gambar 4: Peta environment PlatformIO — 6 runtime environment deploy ke hardware.

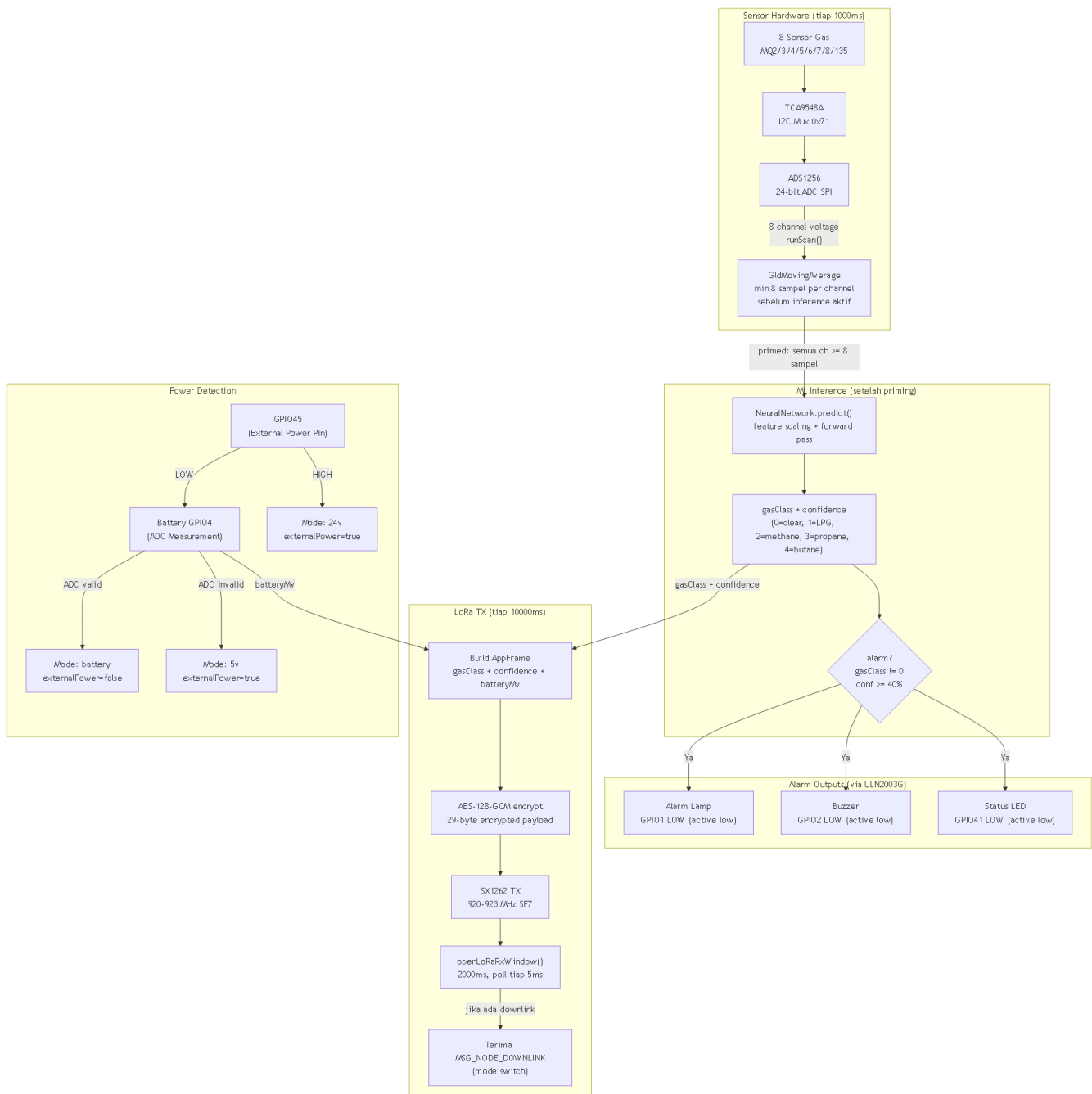
Tabel Fungsi Environment

Env	Role	Fungsi
gld	GLD (ID 0xF001)	GLD unified: inference + dataset + nulling
ch1	CH (ID 0x0064)	CH runtime default, batt threshold normal
ch2	CH (ID 0x0065)	CH runtime, batt threshold = 0
ch3	CH (ID 0x0066)	CH runtime
gw	Gateway (ID 0x006F)	Gateway MESH+WiFi+MQTT bridge ke server

■ **Penting:** 6 environment di atas adalah satu-satunya yang digunakan untuk deploy ke hardware (`firmware/platformio.ini`). Environment `bench/selftest` dan `support/legacy` ada di folder terpisah dan bukan untuk produksi.

4. GLD Node

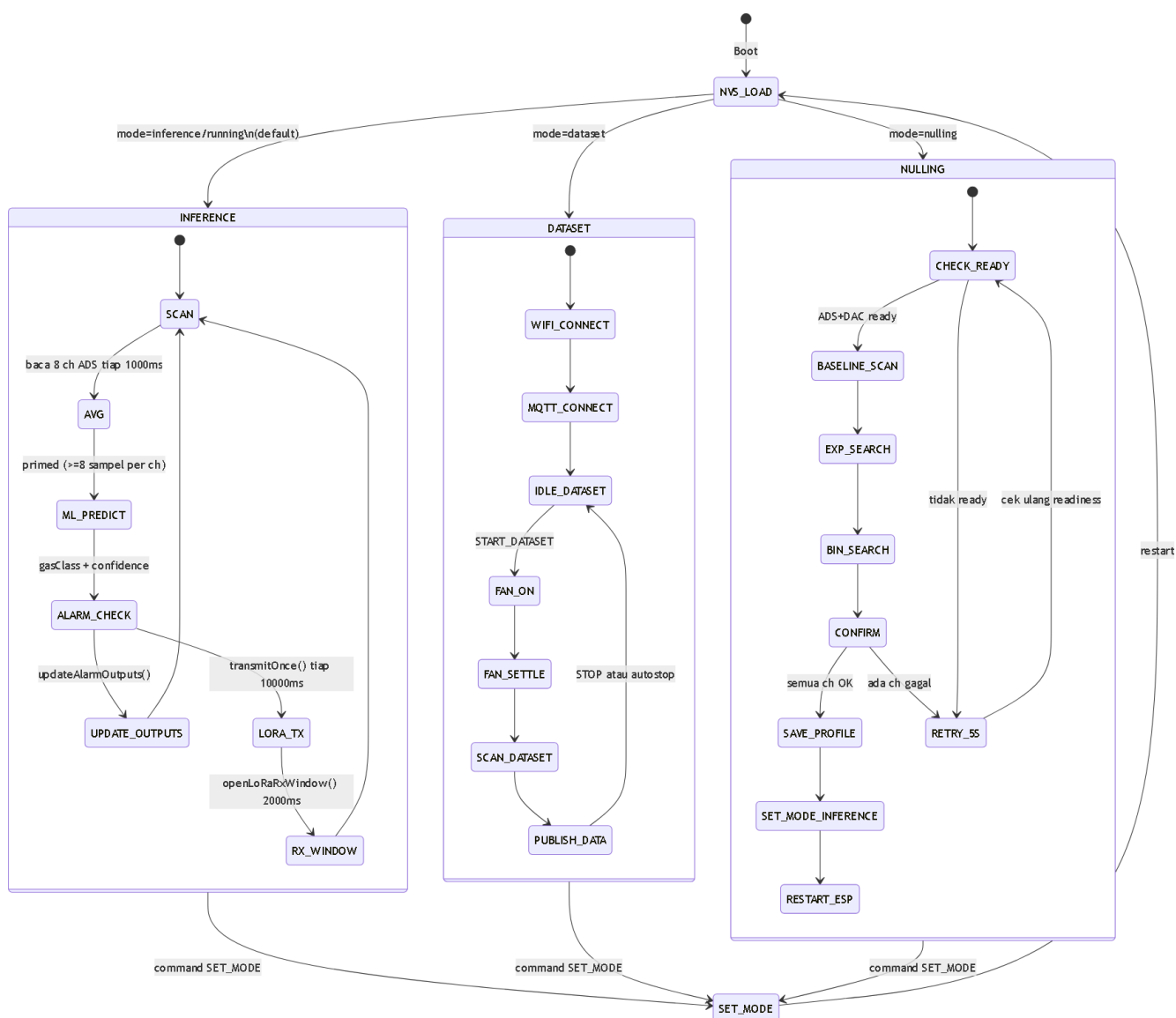
Diagram Internal GLD



Gambar 5: Diagram internal GLD Node berdasarkan source firmware — scan sensor tiap 1000ms, TX LoRa tiap 10000ms.

■ **Catatan:** DC Fan (GPIO42) diinisialisasi LOW dan tidak dikontrol dalam inference mode. Fan hanya aktif di dataset mode saat menerima perintah `START_DATASET` dengan `use_fan_intake=true`.

Diagram Mode GLD



Gambar 6: Diagram mode GLD — inference (dengan fan dan sleep battery), dataset, dan nulling.

■ **Catatan:** Diagram di atas mencerminkan firmware yang ada. DC Fan tidak dikontrol dalam inference — hanya aktif di dataset mode. Sleep setelah RX_WINDOW belum diimplementasikan (loop terus berjalan).

Tabel Pin/Config Penting GLD

Fungsi	Pin/Nilai
SPI SCK/MOSI/MISO	GPIO12/GPIO11/GPIO13
ADS1256 CS/DRDY/SYNC	GPIO47/GPIO10/GPIO18
LoRa CS/RST/BUSY/DIO1	GPIO15/GPIO39/GPIO7/GPIO40 (4D)
LoRa RXEN/TXEN	GPIO5/GPIO6
I2C SDA/SCL	GPIO8/GPIO9
TCA9548A	I2C 0x71
MCP4725 (DAC)	I2C 0x60, range 0-4095
Status LED	GPIO41

Fungsi	Pin/Nilai
Alarm Lamp	GPIO1 (4D) / disabled WROOM
Buzzer	GPIO2 (4D) / disabled WROOM
DC Fan	GPIO42
Battery ADC	GPIO4
24V Power-Good	GPIO45
TPL5110 DONE	GPIO14 (OUTPUT LOW saat init)

WROOM profile overrides: LoRa CS/RST/BUSY/DIO1 → GPIO7/GPIO2/GPIO15/GPIO1; Alarm Lamp/Buzzer → disabled (-1).

Tabel Sensor Order dan MUX

HW Channel	Sensor	TCA Mux Ch	Model Input
0	MQ8	7	0
1	MQ135	6	2
2	MQ3	5	5
3	MQ5	4	3
4	MQ4	3	4
5	MQ7	2	6
6	MQ6	0	1
7	MQ2	1	7

Tabel Command GLD

Command	Parameter	Serial (CDC/UART0)	MQTT	LoRa Downlink
SET_MODE inference	mode=0	Ya	Ya (topic /cmd)	Ya (byte[0]=0x01, byte[1]=0)
SET_MODE dataset	mode=1	Ya	Ya (topic /cmd)	Ya (byte[0]=0x01, byte[1]=1)
SET_MODE nulling	mode=2	Ya	Ya (topic /cmd)	Ya (byte[0]=0x01, byte[1]=2)
SET_MODE running	alias inference	Ya	Ya (topic /cmd)	Tidak
DEBUG_ON	—	Ya	Tidak	Tidak
DEBUG_OFF	—	Ya	Tidak	Tidak
START_DATASET	label, target, dll	Tidak	Ya (topic /dataset)	Tidak
STOP_DATASET	—	Tidak	Ya (topic /dataset)	Tidak

MQTT topic: `gas-leak-detector/F001/cmd` untuk mode; `gas-leak-detector/F001/dataset` untuk dataset.

LoRa Downlink via `MSG_NODE_DOWNLINK`: dikirim dari CH ke GLD setelah CMD diterima dari Server.

Skenario Penggunaan Command

Skenario Operator	Command	Via
Kembali ke monitoring normal setelah kalibrasi atau debug	SET_MODE inference	Serial, MQTT, atau LoRa (remote)
Mulai pengambilan data training sensor	SET_MODE dataset lalu START_DATASET	Serial + MQTT
Kalibrasi sensor setelah pemasangan baru atau relokasi	SET_MODE nulling	Serial, MQTT, atau LoRa (remote)
Debug real-time: lihat log sensor, inference, dan alarm di serial monitor	DEBUG_ON	Serial (USB)
Matikan log debug setelah selesai investigasi	DEBUG_OFF	Serial (USB)
Remote ganti mode GLD dari server tanpa akses fisik	SET_MODE inference/dataset/nulling	Server ke MQTT ke GW ke CH ke GLD
Hentikan sesi dataset sebelum target sampel tercapai	STOP_DATASET	MQTT

Model ML dan Alarm Rule

Alarm rule (dari source):

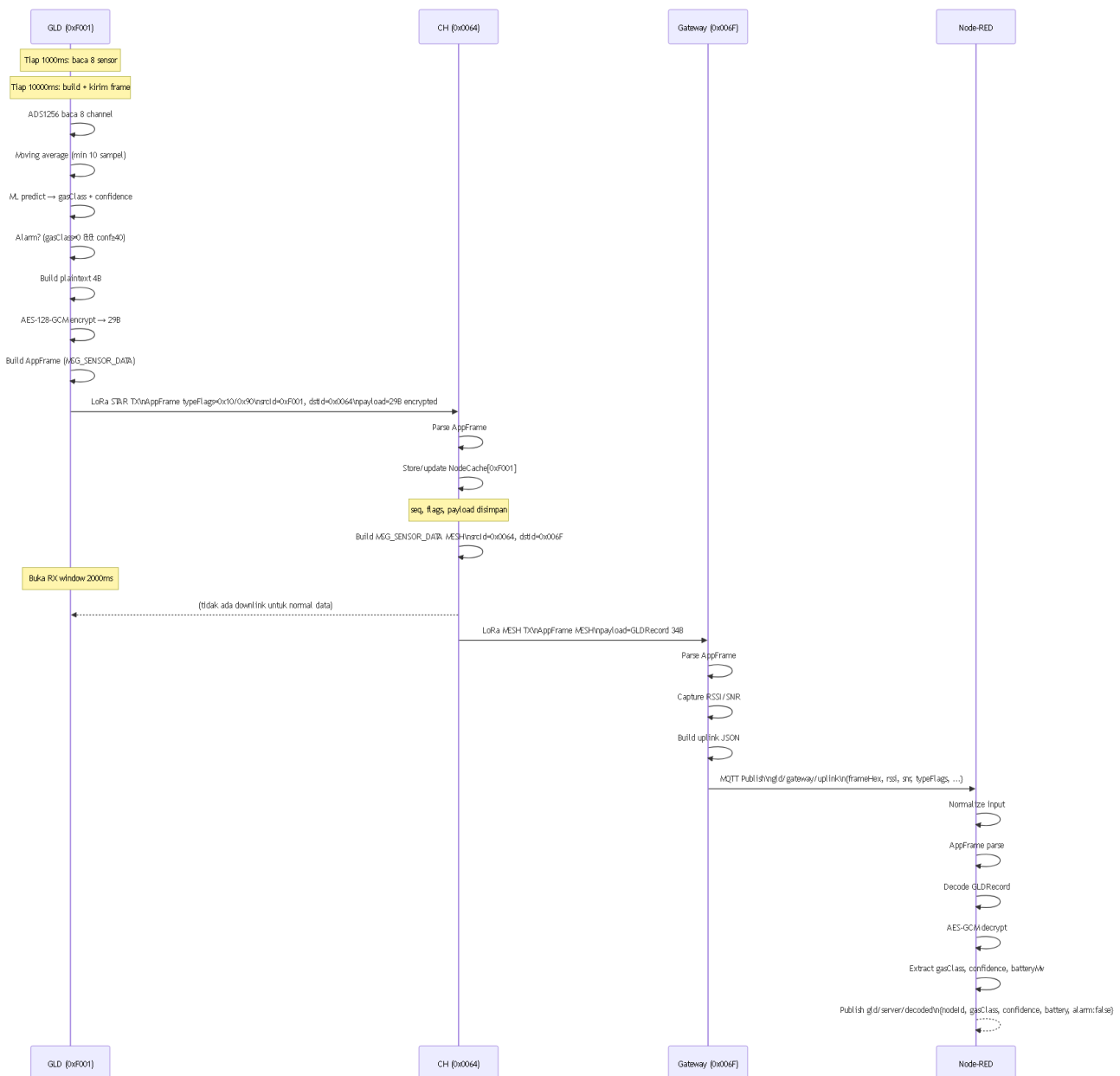
```
alarm = gasClass != clearGas (0) && confidence >= 40
```

Model Class	Protocol Gas Class	Nama Gas
0	0	clearGas
1	1	LPG
2	4	methane
3	2	propane
4	3	butane
lainnya	6	anomaly

■ **Catatan:** Confidence disimpan sebagai `uint8_t(confidenceFloat * 100.0f)`. Nilai valid 0–100.

Normal Data Flow

Sequence Diagram Normal Data Flow



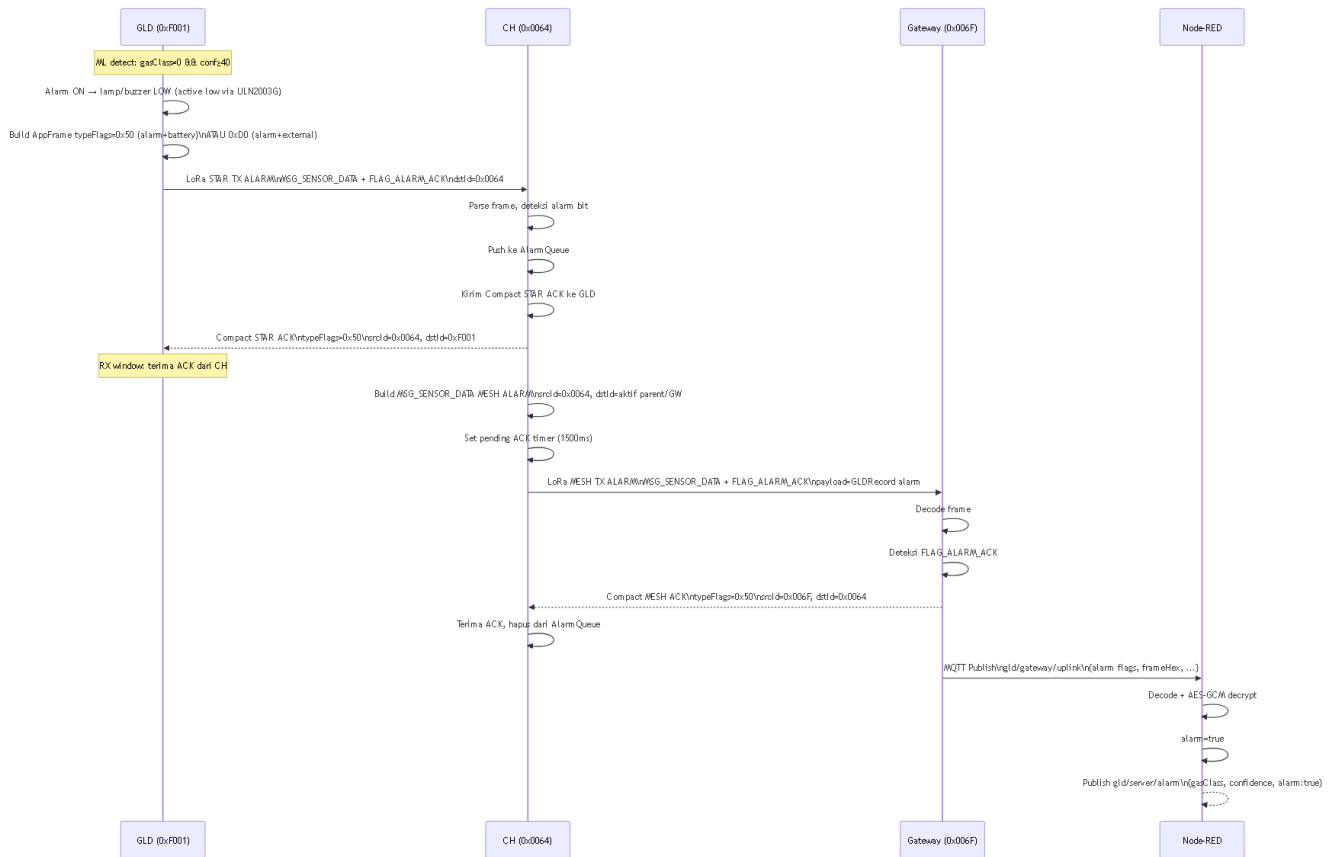
Gambar 13: Sequence diagram alur data normal GLD → CH → Gateway → Node-RED.

Tabel Input/Output Per Tahap — Normal Flow

Tahap	Input	Proses	Output
GLD Scan	Tegangan ADS1256 × 8	Moving avg + ML predict	gasClass, confidence, batteryMv
GLD Encode	4-byte plaintext	AES-128-GCM + AppFrame	29-byte encrypted dalam frame
CH Cache	AppFrame STAR	Parse + NodeCache update	GLDRecord dalam cache
CH MESH TX	NodeCache entry	Build MSG_SENSOR_DATA MESH	AppFrame MESH ke GW
GW Receive	AppFrame MESH	Parse + RSSI/SNR	JSON uplink
GW Publish	JSON uplink	PubSubClient.publish	MQTT gld/gateway/uplink
NR Decode	JSON dengan frameHex	AppFrame + GLD + AES-GCM	gld/server/decoded

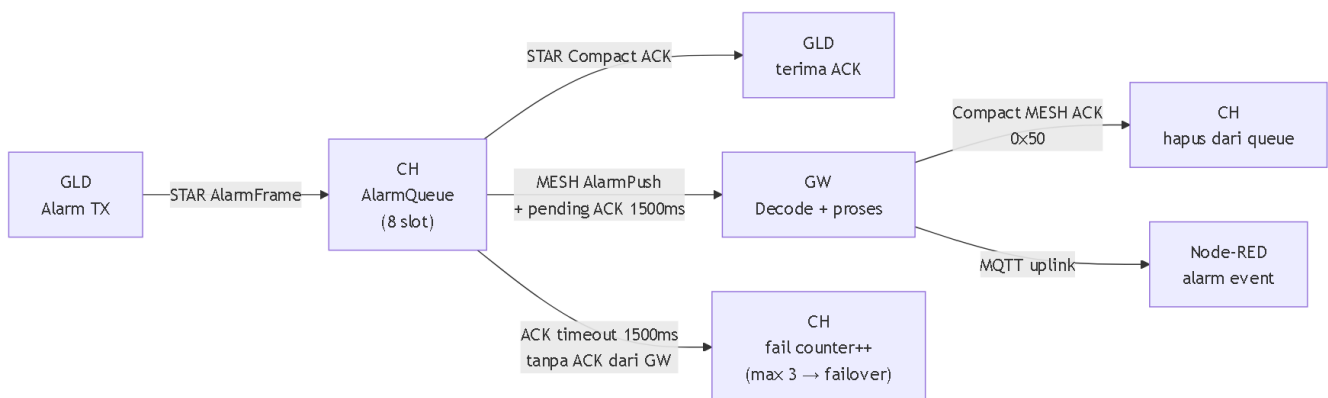
Alarm Flow

Sequence Diagram Alarm



Gambar 14: Sequence diagram alarm flow — dari deteksi GLD hingga server.

Diagram Queue/ACK Alarm



Gambar 15: Mekanisme queue dan ACK alarm CH.

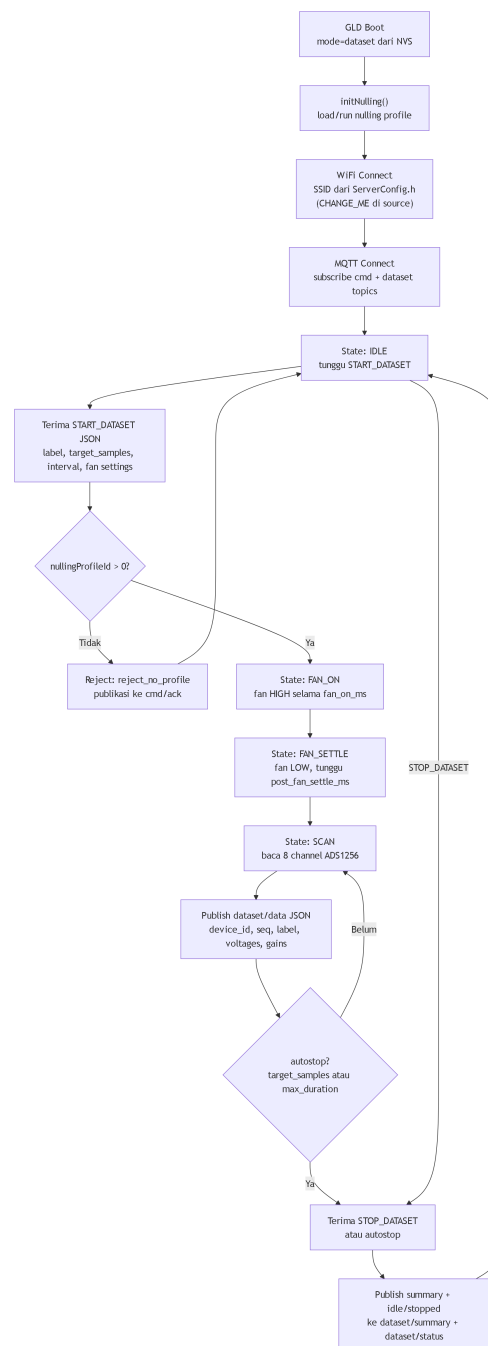
Tabel Alarm Flags

Flag	Nilai	Penggunaan
FLAG_ALARM_ACK	0x40	Bit alarm dalam typeFlags
FLAG_GLD_EXT_POWER	0x80	Bit external power dalam typeFlags
NC_FLAG_ALARM	0x01	Flag alarm dalam GLDRecord/NodeCache
NC_FLAG_EXT_POWER	0x10	Flag external power dalam GLDRecord

■ **Caveat:** Saat ini `checkAlarmAckTimeout()` CH hanya log timeout, hapus item dari `AlarmQueue`, dan increment counter — **tidak retry** alarm frame yang sama. Parent failover terpicu jika counter mencapai threshold 3, dan `RECOVERY` (restart ESP) terpicu jika no-ACK burst mencapai 5.

Dataset Mode Flow

Flowchart Dataset Mode



Gambar 19: Flowchart dataset mode — inialisasi, capture loop, dan autostop.

Tabel MQTT Topics Dataset

Topic	Arah	Isi
gas-leak-detector/F001/cmd	Server → GLD	{"cmd": "SET_MODE", "mode": "..."}
gas-leak-detector/F001/dataset	Server → GLD	{"cmd": "START_DATASET" atau "STOP_DATASET", ...}
gas-leak-detector/F001/dataset/data	GLD → Server	JSON data tiap scan
gas-leak-detector/F001/dataset/status	GLD → Server	Status idle/running/stopped

Topic	Arah	Isi
gas-leak-detector/F001/dataset/summary	GLD → Server	Ringkasan sesi dataset
gas-leak-detector/F001/cmd/ack	GLD → Server	Ack command (termasuk reject)

Tabel Field JSON Dataset Data

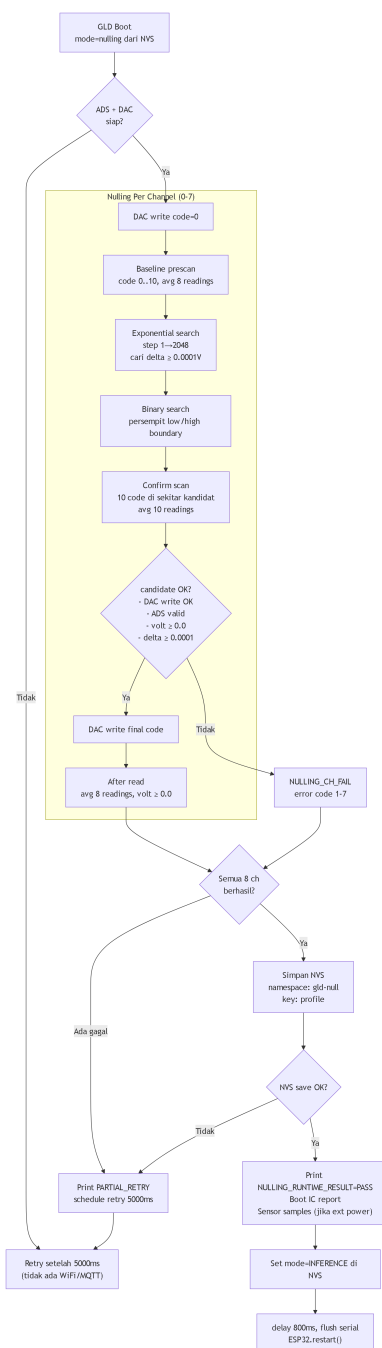
Field	Keterangan
device_id	"F001"
node_id	0xF001 numerik
mode	"DATASET"
seq	Sequence dataset
timestamp_ms	millis()
label	Label dari START_DATASET
nulling_profile_id	Profile ID yang dipakai
sensor_voltage	Array 8 tegangan (float)
sensor_gain	Array 8 gain/status dari pembacaan
feature_order	Nama sensor dalam urutan hardware

Parameter START_DATASET

Field	Default	Keterangan
label	"unknown"	Label untuk data capture
target_samples	0	0 = infinite
sample_interval_ms	1000	Interval antar scan
max_duration_ms	0	0 = infinite
use_fan_intake	true	Gunakan kipas intake
fan_on_ms	1000	Durasi kipas menyala
post_fan_settle_ms	0	Settle setelah kipas mati

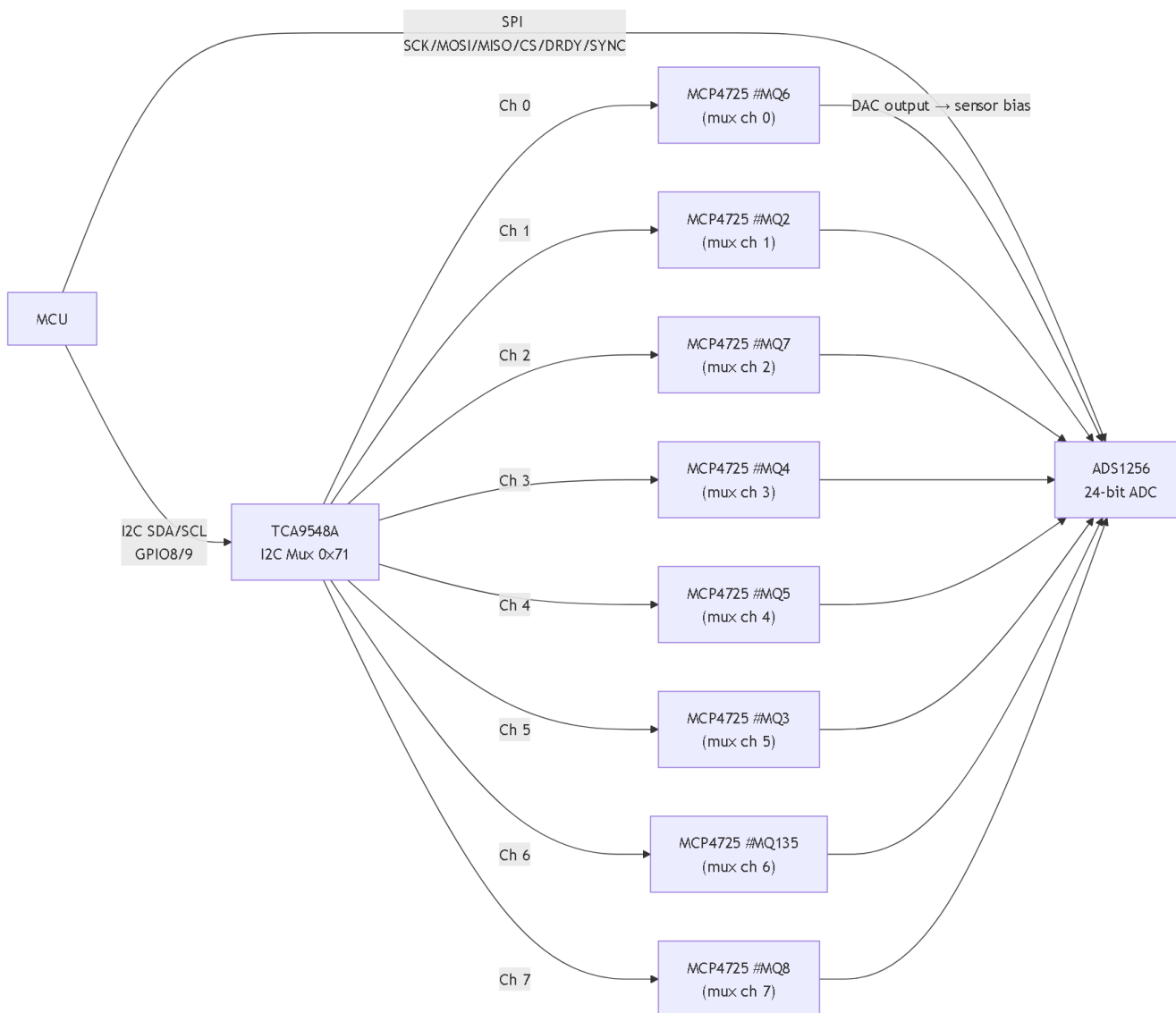
Nulling Mode Flow

Flowchart Nulling Mode



Gambar 20: Flowchart nulling mode — per-channel calibration loop.

Diagram Relasi DAC/MUX/ADS



Gambar 21: Relasi DAC (MCP4725 via TCA9548A) dan ADC (ADS1256) dalam nulling.

Tabel Threshold/Count/Timing Nulling

Konstanta	Nilai	Keterangan
Average count	8	Jumlah sampel per rata-rata
Confirm count	10	Jumlah sampel per kandidat confirm
Baseline prescan max code	10	Kode DAC 0..10 untuk prescan
Exponential initial step	1	Step awal exponential search
Exponential max step	2048	Step maksimum sebelum berhenti
Confirm window	10 DAC codes	Area konfirmasi di sekitar kandidat
Settle delay	5 ms	Jeda setelah DAC write
Threshold	0.0001 V	Delta minimum yang diterima
Minimum final voltage	0.0 V	Tegangan akhir harus $\geq$ ini

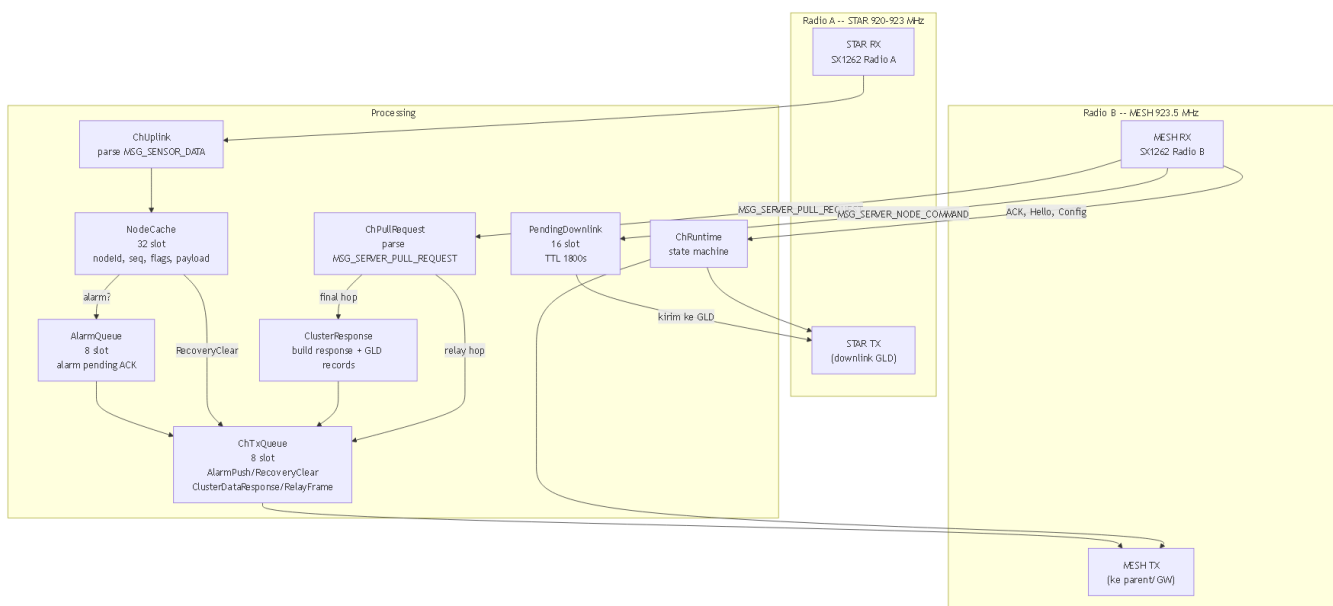
Nulling Error Codes

Code	Alasan Gagal
1	dac_zero_write_failed
2	baseline_no_valid_samples
3	exponential_range_not_found
4	confirm_failed
5	dac_final_write_failed
6	after_read_invalid
7	after_voltage_negative

■ **Catatan:** Nulling mode berjalan **offline** — tidak ada WiFi, MQTT, atau LoRa. Setelah sukses, firmware otomatis mengubah mode ke **inference** dan restart ESP.

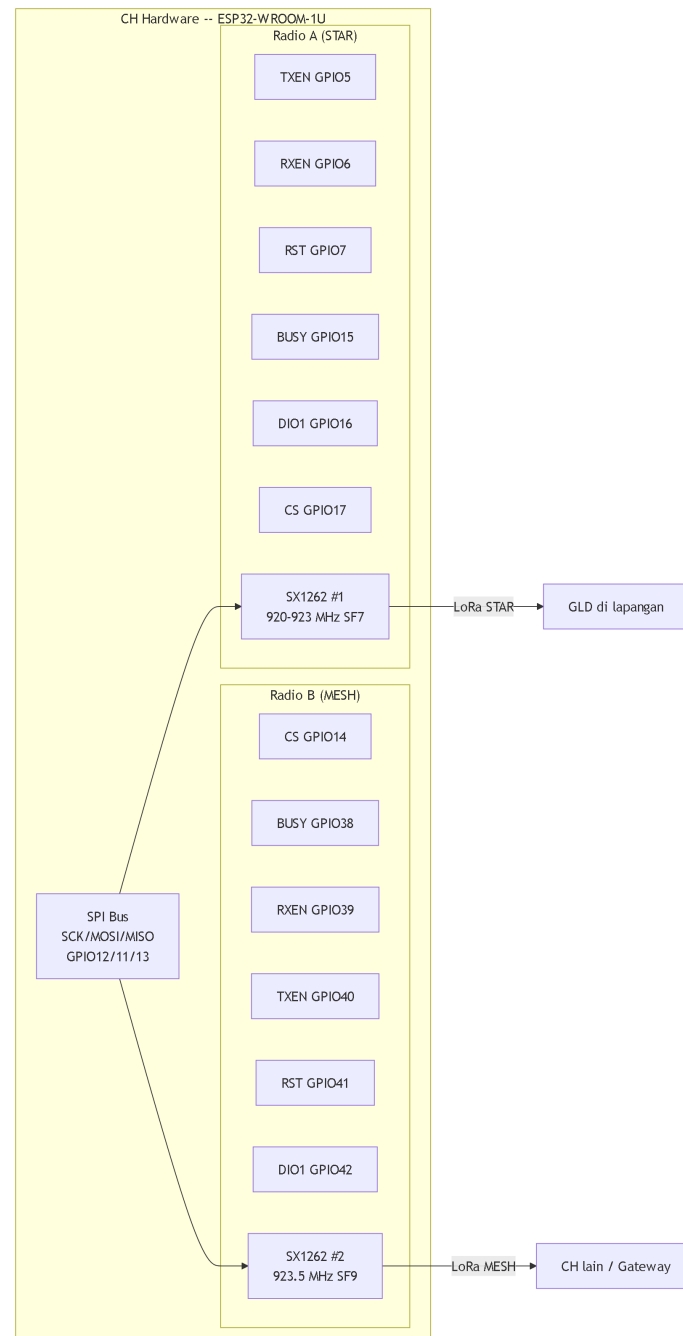
5. Cluster Head

Diagram Internal CH



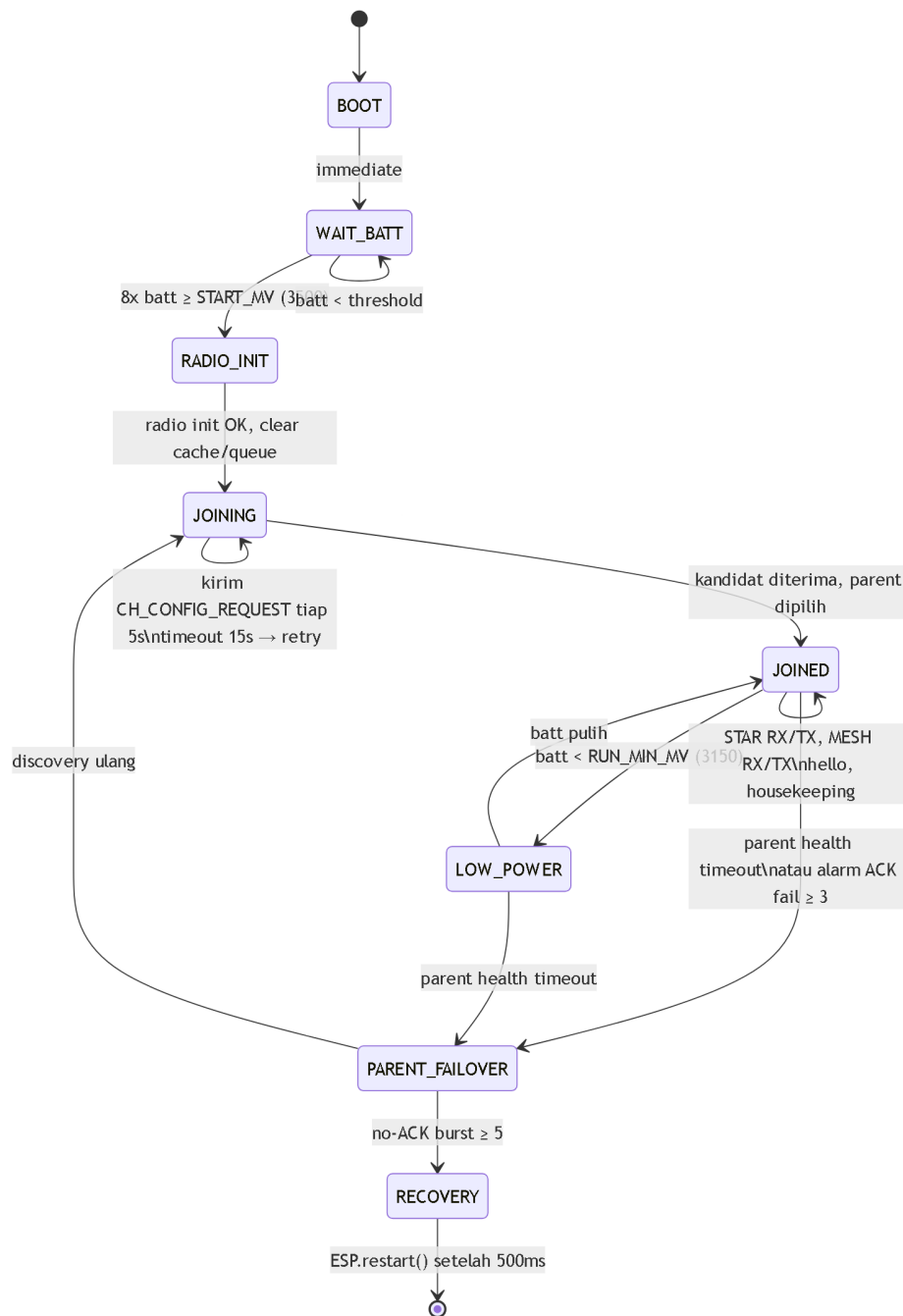
Gambar 7: Diagram internal Cluster Head — dual radio, cache, queue, dan routing.

Diagram Dual Radio CH



Gambar 8: Dual radio CH — Radio A untuk STAR (GLD), Radio B untuk MESH (backbone).

Diagram State Machine CH



Gambar 9: State machine Cluster Head — dari boot hingga operasi normal.

Tabel CH Identitas per Environment

Env	CH ID	Hex	Battery Thresholds
ch1	100	0x0064	Start: 3500mV, Run: 3150mV, Critical: 3100mV
ch2	101	0x0065	Semua = 0 (bench, batt tidak diperiksa)
ch3	102	0x0066	Start: 3500mV, Run: 3150mV, Critical: 3100mV
Gateway (root)	111	0x006F	—

Tabel Cache/Queue/Kapasitas CH

Komponen	Kapasitas	TTL / Timeout
NodeCache	32 slot	Stale: 300s, Expire: 3600s
AlarmQueue	8 slot	ACK timeout: 1500ms
MESH TX Queue	8 slot	—
Pending Downlink	16 slot	TTL: 1800000ms (30 menit)
Parent Candidates	8 slot	—

Perilaku CH_HELLO dan Parent Discovery

CH_HELLO payload (11 byte):

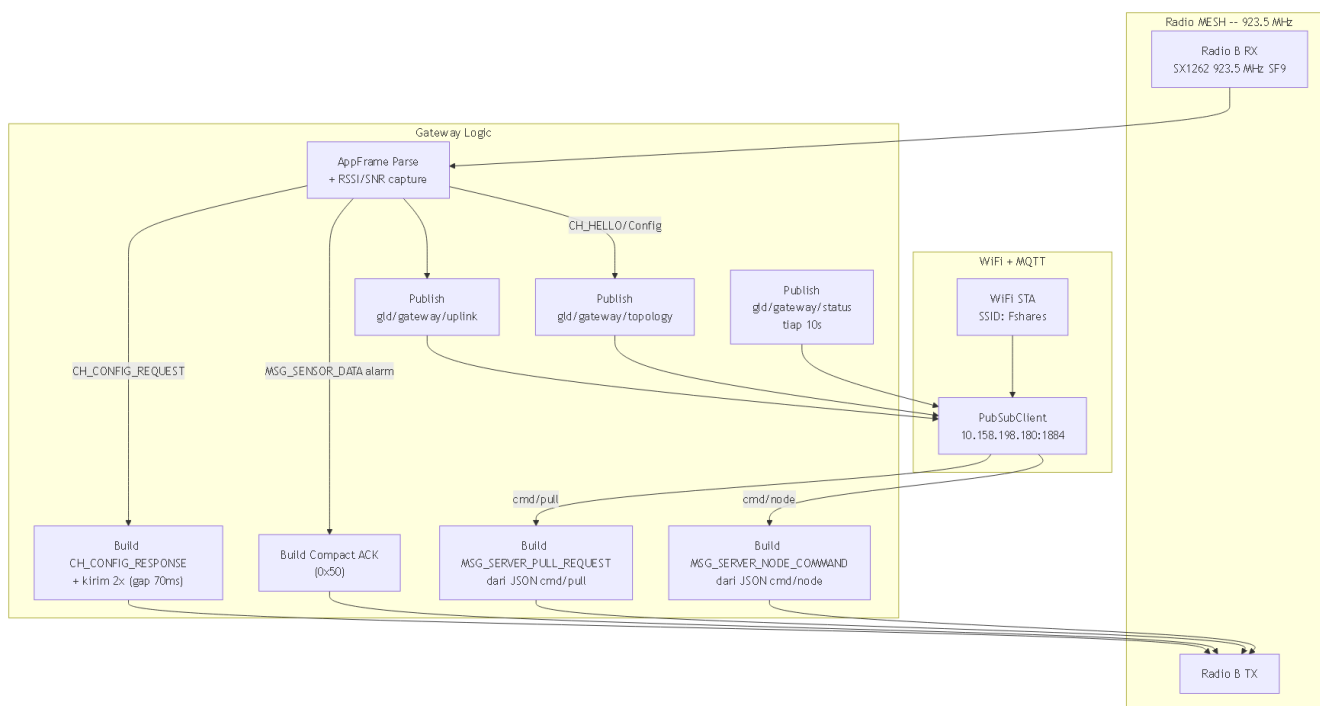
Offset	Field	Ukuran
0–1	CH ID	2
2–3	Parent ID	2
4–5	Battery mV	2
6–7	Uptime sec (low 16-bit)	2
8	Mesh depth	1
9–10	Alternate parent ID	2

Discovery timing:

Parameter	Nilai
Joining timeout	15000 ms
Config request interval	5000 ms
Response base delay	200 ms
Response slot gap	280 ms
Slot count	16
Jitter formula	$20 + (\text{CH_ID} \& 0x000F) * 30 + (\text{millis}() \% 40) \text{ ms}$

6. Gateway

Diagram Internal Gateway



Gambar 10: Diagram internal Gateway — MESH RX/TX dan MQTT bridge.

Tabel MQTT Topics Gateway

Topic	Arah	Isi Payload
gld/gateway/uplink	GW → Server	JSON wrapper raw MESH frame (frameHex, rssi, snr, parse, typeFlags, ...)
gld/gateway/topology	GW → Server	JSON topology dari CH_HELLO, CH_CONFIG_REQUEST/RESPONSE
gld/gateway/status	GW → Server	JSON status: gatewayId, wifi, mqtt, meshReady, ip
gld/gateway/cmd/#	Server → GW	Wildcard subscription
gld/gateway/cmd/pull	Server → GW	JSON pull command: requestId + hopList
gld/gateway/cmd/node	Server → GW	JSON node command: cluster, node, id, ttl, hex

Tabel Config WiFi/MQTT Gateway

Parameter	Nilai
WiFi SSID	Fshares
WiFi Password	kayabiasa
MQTT Host	10.158.198.180
MQTT Port	1884
MQTT User	deviot
MQTT Pass	deviot
Gateway ID	0x006F

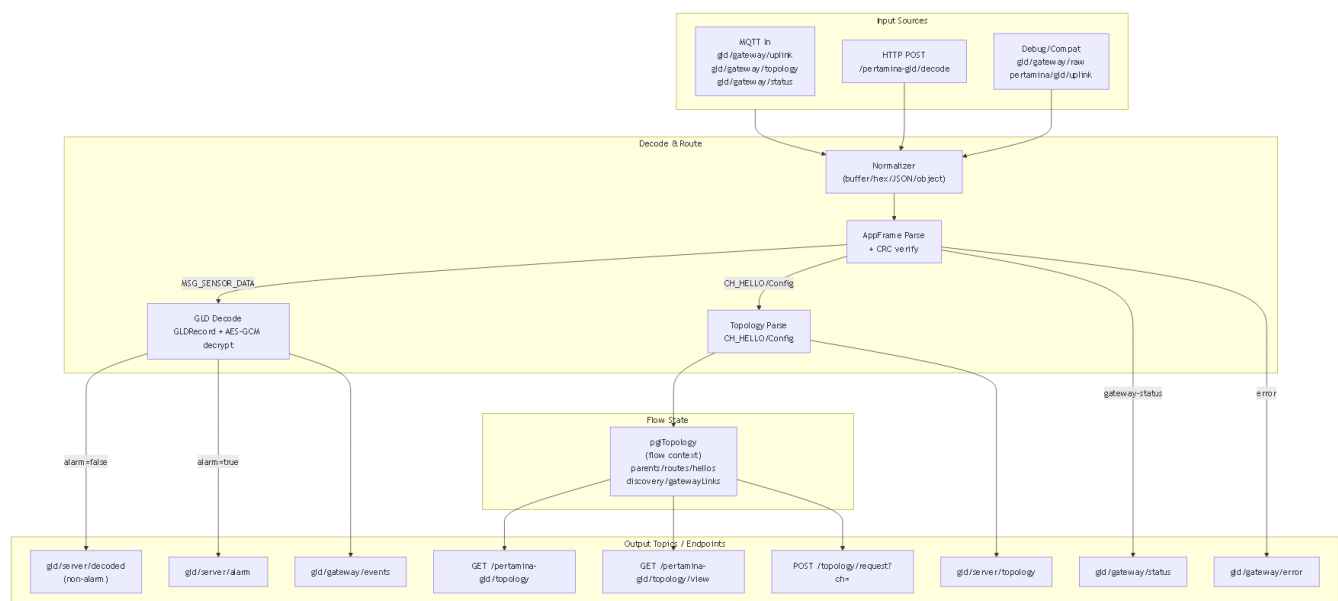
Parameter	Nilai
Topic Root	gld/gateway
MQTT Client ID	pgl-gateway-006F-<MAC>
WiFi Retry	5000 ms
MQTT Retry	3000 ms
Status Interval	10000 ms
Config Response Repeat	2x, gap 70 ms

Gateway Uplink JSON Fields

Field	Kapan Ada
source = "gateway"	Selalu
gatewayId	Selalu
frameHex	Selalu
frameLen	Selalu
rssi, snr	Selalu
parseStatus	Selalu
typeFlags, msgType, srcId, dstId, seq, payloadLen	AppFrame parse OK
topology (object embed)	Parse OK + MSG_CH_HELLO payload ≥8 byte

7. Server / Node-RED

Diagram Node-RED Logical Flow



Gambar 11: Logical flow Node-RED — input, decode, routing, dan output.

Tabel HTTP Endpoints Node-RED

Endpoint	Method	Fungsi
/pertamina-gld/decode	POST	Manual decode frame hex
/pertamina-gld/topology	GET	JSON topology: nodes, edges, routes, discovery
/pertamina-gld/topology/view	GET	HTML UI topology (auto-refresh 1000ms)
/pertamina-gld/topology/reset	POST	Reset state: parents, discovery, hellos, routes
/pertamina-gld/topology/request?ch=<id>	POST	Publish pull request ke cmd/pull jika route ada
/pertamina-gld/topology/delete?ch=<id>	POST	Hapus CH dari semua map topology

Tabel Topology State (pglTopology)

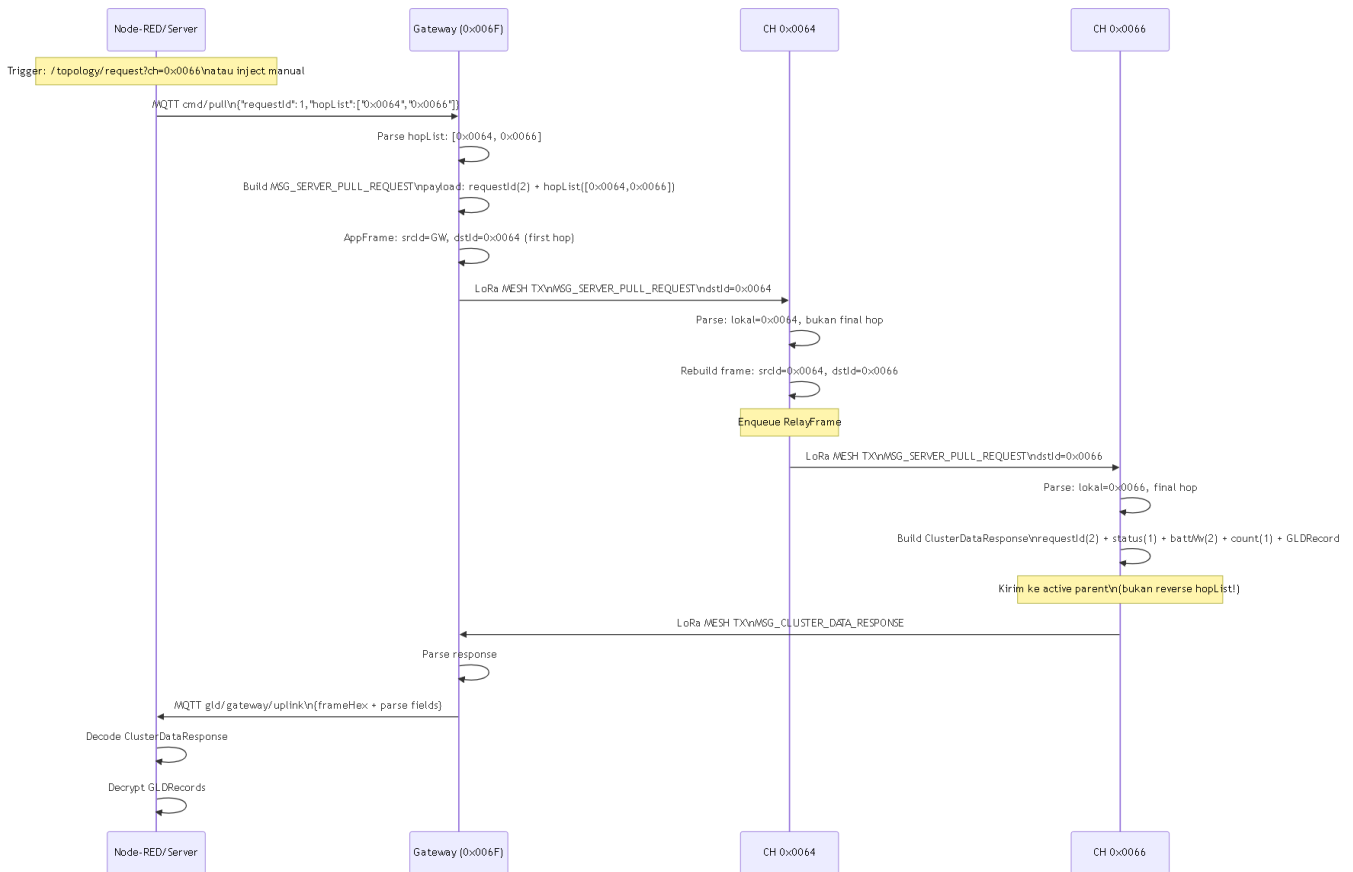
Field	Isi	TTL Default
parents	Installed CH parent (dari CH_HELLO)	900000 ms
discovery	CH_CONFIG discovery candidates	420000 ms
gatewayLinks	Gateway-heard RSSI/SNR dari CH_CONFIG	420000 ms
hellos	Latest CH_HELLO per CH	—
routes	Computed hop list Gateway→CH	—

Decode Output Fields (GLD Event)

Field	Keterangan	
ok	true jika decode berhasil	
kind	gld-event	
receivedAt	ISO timestamp	
nodeId, nodeIdHex	GLD ID	
seq	GLD record sequence	
flags	GLDRecord flags byte	
alarm	bit 0x01 dari flags	
externalPower	bit 0x10 dari flags	
testDevice	node ID dalam range 0xF000..0xFEFF	
payloadLen, payloadHex	encrypted payload	
dedupKey	`<cluster>:<node>:<seq>:<alarm\`	normal>`
decryptOk	AES-GCM decrypt berhasil	
gasClass, gasName	hasil decrypt	
confidence	0–100	
batteryMv	mV baterai, 0xFFFF = invalid	

Server Pull Flow

Sequence Diagram Server Pull



Gambar 16: Sequence diagram server pull flow — hopList routing hingga respons.

Diagram HopList

Server kirim: hopList = [0x0064, 0x0066]

Step 1: GW → CH 0x0064 (first hop)

AppFrame: srcId=GW(0x006F), dstId=0x0064

Payload: requestId + [0x0064, 0x0066]

Step 2: CH 0x0064 relay → CH 0x0066

CH 0x0064 baca payload, cari posisi lokal (index 0)

bukan final hop → relay ke next: dstId=0x0066

AppFrame: srcId=0x0064, dstId=0x0066

Payload: requestId + [0x0064, 0x0066] (sama)

Step 3: CH 0x0066 final → kirim respons ke active parent

Final hop → build ClusterDataResponse

Kirim ke runtimeConfig.meshDstId (parent aktif, bukan reverse)

Tabel Data Status Respons Pull

Nilai	Nama	Arti
0x00	DataOk	Data tersedia dan valid
0x01	DataEmpty	Cache CH kosong, tidak ada GLD
0x02	DataNotAvail	Data tidak tersedia (belum ada data)
0x03	DataStale	Data ada tapi terlalu lama (>300s)
0x04	DataBusy	CH sedang sibuk
0x05	DataInvalid	Data tidak valid

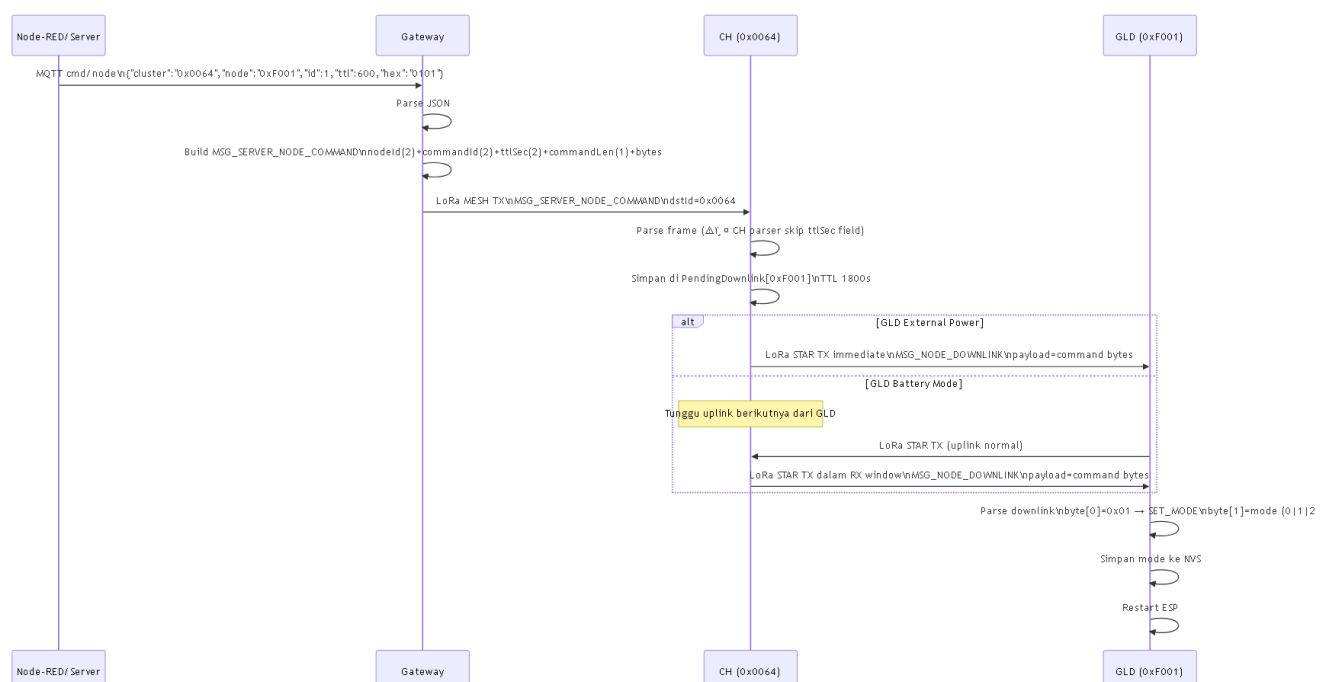
ClusterDataResponse Payload

Offset	Field	Ukuran
0–1	requestId	2
2	data status	1
3–4	CH battery mV	2
5	record count	1
6..	GLDRecords (34 byte per record)	variable

■ **Kapasitas:** MESH payload max 80 byte. Header 6 byte + 2 × 34 byte = 74 byte. Maksimal 2 GLDRecord per respons.

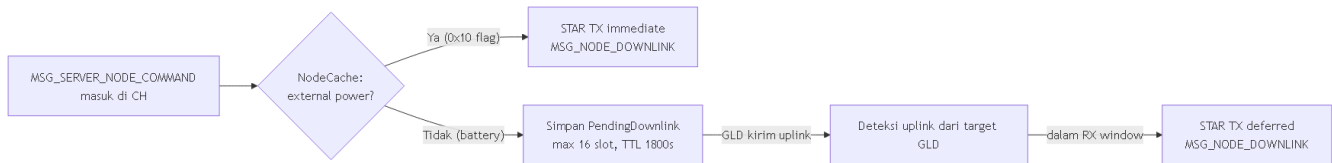
Downlink Command Flow

Sequence Diagram Downlink



Gambar 17: Sequence diagram downlink command flow.

Immediate vs Deferred Downlink



Gambar 18: Immediate downlink (external power) vs deferred downlink (battery mode).

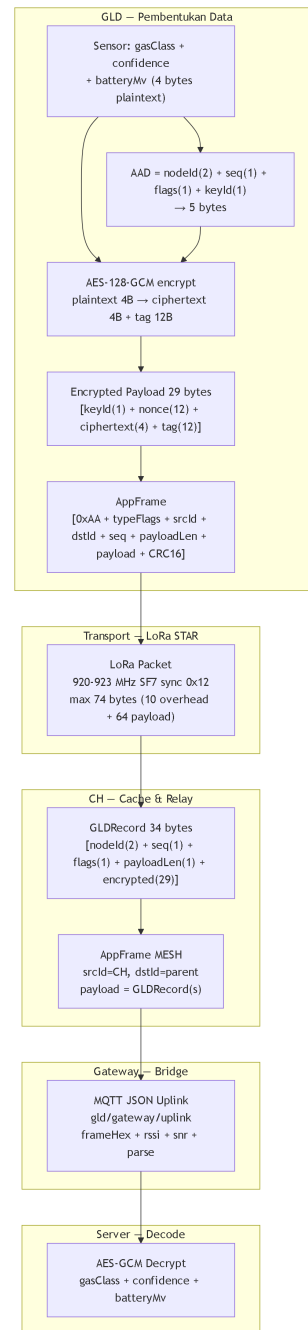
GLD Downlink Parser

Byte	Field	Nilai Saat Ini
0	Command type	0x01 = SET_MODE
1	Mode value	0 = inference, 1 = dataset, 2 = nulling

■ **Caveat TTL Mismatch:** Gateway build payload `nodeId(2) + commandId(2) + ttlSec(2) + commandLen(1) + commandBytes`, sedangkan CH parser membaca `nodeId(2) + commandId(2) + commandLen(1) + commandBytes`. Field `ttlSec` dari Gateway dilewati CH dan menempa `commandLen`, sehingga perintah downlink saat ini **tidak ter-align**. Ini adalah mismatch yang ada di source saat ini.

8. Protocol Visual Map

Diagram Lapisan Protocol



Gambar 12: Lapisan protocol dari data sensor hingga dekripsi di server.

Tabel AppFrame Field

Offset	Field	Ukuran	Keterangan
0	Magic 0xAA	1	Identifikasi frame
1	typeFlags	1	Type (6-bit) + Flags (2-bit)
2–3	srcId	2	Source node ID (big-endian)
4–5	dstId	2	Destination node ID (big-endian)
6	seq	1	Sender sequence counter

Offset	Field	Ukuran	Keterangan
7	payloadLen	1	Panjang payload (0–64 STAR, 0–80 MESH)
8..	payload	N	Isi sesuai message type
akhir	CRC16-CCITT-FALSE	2	Over header + payload

Tabel Message Types

Hex	Nama	Penggunaan
0x10	MSG_SENSOR_DATA	GLD uplink, alarm push, recovery clear, compact ACK reuse
0x14	MSG_NODE_DOWNLINK	CH → GLD downlink command
0x30	MSG_SERVER_PULL_REQUEST	Server → GW → CH: pull data request
0x31	MSG_CLUSTER_DATA_RESPONSE	CH → GW: pull data response
0x32	MSG_SERVER_NODE_COMMAND	GW → CH: command ke GLD tertentu
0x33	MSG_CH_HELLO	CH topology/liveness announcement
0x34	MSG_CH_CONFIG_REQUEST	CH discovery broadcast
0x35	MSG_CH_CONFIG_RESPONSE	CH/GW balasan discovery

Tabel typeFlags GLD

Name	Nilai	Kondisi
Normal + Battery	0x10	gasClass=0 atau conf<40, baterai
Normal + External	0x90	gasClass=0 atau conf<40, external power
Alarm + Battery	0x50	alarm aktif, baterai
Alarm + External	0xD0	alarm aktif, external power
Compact Alarm ACK	0x50	ACK dari CH/GW ke pengirim alarm

Diagram GLD Payload — 4 Byte Plaintext dan 29 Byte Encrypted

PLAINTEXT (4 bytes):



ENCRYPTED PAYLOAD (29 bytes):



AAD (5 bytes, tidak dikirim tapi dipakai untuk enkripsi/validasi):



Diagram GLDRecord (34 byte) — Unit Cache dan Respons CH

GLDRecord (34 bytes):



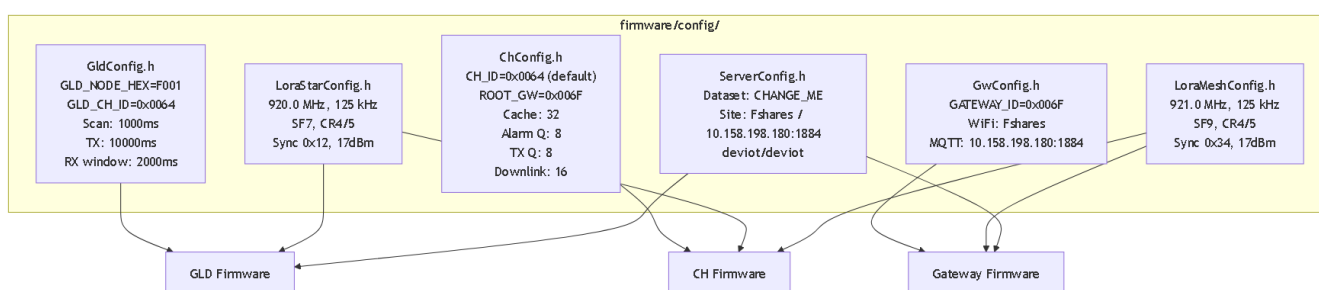
Record flags:

0x01 = alarm

0x10 = external power

9. Konfigurasi Penting

Visual Config Map Per File



Gambar 22: Peta config file dan dependency ke firmware.

Tabel Node IDs

Node	ID (hex)	ID (desimal)	Peran
GLD	0xF001	61441	Sensor gas node
CH1	0x0064	100	Cluster Head 1
CH2	0x0065	101	Cluster Head 2
CH3	0x0066	102	Cluster Head 3
Gateway	0x006F	111	Root Gateway/MESH
Broadcast	0xFFFF	65535	Broadcast ke semua

Tabel Radio Config STAR/MESH

Parameter	STAR (GLD↔CH)	MESH (CH↔GW)
Frekuensi	920.0 MHz	921.0 MHz
Bandwidth	125 kHz	125 kHz

Parameter	STAR (GLD↔CH)	MESH (CH↔GW)
Spreading Factor	SF7	SF9
Coding Rate	4/5	4/5
Sync Word	0x12	0x34
TX Power	17 dBm	17 dBm
Preamble	8	8
SPI Clock	2 MHz	2 MHz
TCXO	1.6V (fallback 0.0V)	1.6V (fallback 0.0V)
Max Payload	64 bytes	80 bytes

Tabel Timing Penting

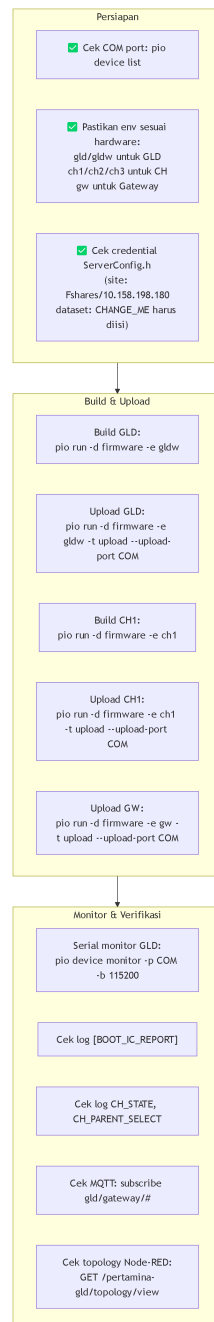
Parameter	Nilai	Komponen
GLD scan interval	1000 ms	GLD
GLD TX interval	10000 ms	GLD
GLD RX window	2000 ms	GLD
GLD WiFi timeout	15000 ms	GLD
GLD MQTT retry	3000 ms	GLD
CH joining timeout	15000 ms	CH
CH config request interval	5000 ms	CH
CH hello interval	300000 ms	CH
CH alarm ACK timeout	1500 ms	CH
CH pending downlink TTL	1800000 ms	CH
CH parent health timeout	180000 ms	CH
CH parent min dwell	300000 ms	CH
CH housekeeping	60000 ms	CH
CH route verify	600000 ms	CH
GW status interval	10000 ms	Gateway
GW WiFi retry	5000 ms	Gateway
GW MQTT retry	3000 ms	Gateway

Tabel Battery Thresholds

Threshold	env:ch1/ch3	env:ch2	Kondisi
Start (BATT_START_MV)	3500 mV	0 (disabled)	Harus terpenuhi 8x berturut sebelum radio init
Run Min (BATT_RUN_MIN_MV)	3150 mV	0 (disabled)	Di bawah ini → LOW_POWER state
Critical (BATT_CRITICAL_MV)	3100 mV	0 (disabled)	TX diblokir di LOW_POWER

10. Operasi Lapangan

Checklist Operasi Visual



Gambar 23: Checklist operasi lapangan.

Command Build/Upload

```
# Cek COM port
pio device list
```

```
# Build GLD (board WROOM, env gldw)
pio run -d firmware -e gldw
```

```
# Upload GLD ke COM9 (contoh)
```

```

pio run -d firmware -e gldw -t upload --upload-port COM9

# Build + Upload CH1
pio run -d firmware -e ch1 -t upload --upload-port COM3

# Build + Upload CH2
pio run -d firmware -e ch2 -t upload --upload-port COM5

# Build + Upload Gateway
pio run -d firmware -e gw -t upload --upload-port COM38

# Serial monitor GLD
pio device monitor -p COM9 -b 115200

# Serial monitor CH
pio device monitor -p COM3 -b 115200

```

■ **Catatan:** COM port di atas adalah contoh dari sesi bench sebelumnya. Selalu cek ulang dengan `pio device list` sebelum upload.

Cara Membaca Log Serial GLD

Log Token	Arti
[BOOT_IC_REPORT]	Tabel IC check setelah boot
[BOOT_SENSOR_SAMPLES]	5 baris sampel sensor (external power)
MODE_READY	Mode siap dieksekusi
NULLING_SERVICE_START	Nulling mulai
NULLING_CH_OK / NULLING_CH_FAIL	Per-channel nulling result
NULLING_RUNTIME_RESULT=PASS	Semua channel sukses
PARTIAL_RETRY	Ada channel gagal, retry

Cara Membaca Log Serial CH

Log Token	Arti
CH_STATE	Transisi state machine
CH_PARENT_SELECT	Parent terpilih
CH_STAR_RX	Terima frame dari GLD
CH_STAR_PROCESS	Proses + update cache
CH_MESH_TX_KIND	Frame MESH yang dikirim
CH_ALARM_ACK_RECV	Terima ACK alarm
CH_ALARM_ACK_TIMEOUT	Timeout ACK alarm
CH_HELLO_TX	Kirim CH_HELLO
CH_CACHE_SUMMARY	Ringkasan isi NodeCache

Cara Membaca Log Serial Gateway

Log Token	Arti
parseStatus di MQTT JSON	Decode result frame MESH
rssI, snr di MQTT JSON	Kualitas link MESH
frameHex	Raw frame untuk debugging
topology object	CH_HELLO embedded

11. Verifikasi dan Test

Diagram Test Coverage



Gambar 24: Coverage test — host-level dan bench self-test.

Tabel Test Command

Command	Platform	Expected
<code>python firmware/tests/run_tests.py</code>	Host (Python)	27/27 tests passed
<code>pio run -d firmware/bench -e gld_sensor_selftest_esp32s3</code>	ESP32-S3 bench	Sensor read OK
<code>pio run -d firmware/bench -e gld_nulling_selftest_esp32s3</code>	ESP32-S3 bench	Nulling OK
<code>pio run -d firmware/bench -e gld_lora_tx_selftest_esp32s3</code>	ESP32-S3 bench	LoRa TX OK
<code>pio run -d firmware/bench -e ch_star_rx_selftest_esp32s3</code>	ESP32-S3 bench	STAR RX parse OK

Tabel Status Test (Yang Boleh Diklaim)

Area	Status
Protocol constants (source)	■ Terimplementasi, ter-guard test host
Firmware versions (source)	■ Terimplementasi, ter-guard test host
Active env config (platformio.ini)	■ Ter-guard test host
GLD board pin macros	■ Ter-guard test host
GLD unified scaffolds	■ Ter-guard test host
Node-RED pull hopList contract	■ Ter-guard test host
GLD sensor read	Source ada, bench test terpisah

Area	Status
Nulling calibration	Source ada, bench test terpisah
LoRa STAR TX	Source ada, bench test terpisah
CH STAR RX parse	Source ada, bench test terpisah
End-to-end GLD→Server dengan AES-GCM	Source + Node-RED ada, belum ada test otomatis E2E
Multi-hop pull via hopList	Source ada, belum ada test integrasi otomatis
Downlink command (TTL mismatch)	Source ada caveat, tidak ter-guard test

12. Known Caveats / Risiko

Area	Caveat	Dampak	Status di Source	Rekomendasi
Downlink TTL mismatch	Gateway build nodeId+commandId+ttlSec+commandLen+bytes, CH parser skip ttlSec sehingga commandLen salah offset	Downlink command dari server ke GLD tidak berfungsi via remote	Terdokumentasi eksplisit di <code>ch-gw/final_design.md</code> dan <code>gld-ch/payload-contract.draft.md</code>	Align satu sisi (tambah ttlSec parsing di CH atau hapus dari GW) sebelum live deploy
Dataset WiFi credential	PGL_SERVER_DATASET_WIFI_SSID/PASSWORD dan MQTT_HOST di ServerConfig.h bernilai "CHANGE_ME"	Dataset mode tidak bisa konek WiFi/MQTT sampai credential diisi	Terlihat di <code>firmware/config/ServerConfig.h</code>	Isi dengan credential prod sebelum build dataset
Alarm ACK tidak retry	Saat ACK timeout (1500ms), CH hanya log dan hapus alarm dari queue — tidak kirim ulang	Alarm bisa hilang jika link MESH sementara drop	Terdokumentasi di <code>ch/final_design.md</code>	Implementasikan alarm retry logic jika reliabilitas diutamakan
Dataset flow JSON mismatch	<code>pertamina-gld-dataset.flow.json</code> expect field <code>ch</code> , <code>gain</code> , <code>ok</code> , <code>nodeId</code> , <code>ts_ms</code> , <code>profileId</code> — sedangkan GLD unified kirim <code>sensor_voltage</code> , <code>sensor_gain</code> , <code>node_id</code> , <code>timestamp_ms</code> , <code>nulling_profile_id</code>	Dataset recorder dari flow lama tidak kompatibel dengan GLD unified	Terdokumentasi di <code>server/final_design.md</code>	Gunakan <code>gld_dataset_recorder.py</code> yang cocok, atau update flow dataset
MQTT buffer limit	GLD dan GW MQTT buffer 1024 bytes	Payload besar (dataset JSON + 8 voltage) mendekati limit	Source: <code>GLD_MQTT_BUFFER_SIZE = 1024</code>	Verifikasi ukuran payload dataset di lapangan
NodeCache expire	Entry GLD expire setelah 3600s tanpa data	CH tidak kirim data GLD yang lama ke server sampai GLD kirim lagi	Source: <code>CACHE_EXPIRE_MS = 3600000</code>	Normal behavior, dokumentasikan ke operator
Gateway parent di CH	CH mem-boot ulang fresh setiap restart dan Gateway link harus fresh (NVS Gateway parent di-clear saat boot)	Perlu waktu re-discovery setelah restart CH	Source: <code>ch/final_design.md</code>	Biarkan joining timeout 15s untuk stabilisasi

Area	Caveat	Dampak	Status di Source	Rekomendasi
Default test AES key	Node-RED default key 000102030405060708090A0B0C0D0E0F via env GLD_AES128_KEY_HEX	Jika env tidak diset, server memakai test key yang sama dengan GLD	Source: server/final_design.md	Set GLD_AES128_KEY_HEX env di prod Node-RED dan GLD

13. Appendix

A. Visual Glossary

Term	Definisi
AppFrame	Format frame biner sistem: magic + typeFlags + srcId + dstId + seq + payloadLen + payload + CRC16
AES-128-GCM	Enkripsi authenticated dengan key 128-bit, nonce 12 byte, tag 12 byte
AAD	Additional Authenticated Data — 5 byte untuk validasi enkripsi tanpa decrypt
CH	Cluster Head — node relay antara GLD dan Gateway
GLD	Gas Leak Detector — node sensor gas utama
STAR	Topologi radio GLD↔CH — satu GLD ke satu CH
MESH	Topologi radio CH↔CH↔Gateway — multi-hop backbone
NodeCache	Cache GLD record di CH — 32 slot, stale 300s, expire 3600s
AlarmQueue	Queue alarm GLD pending ACK di CH — 8 slot, ACK timeout 1500ms
GLDRecord	Unit data GLD 34 byte: nodeId+seq+flags+payloadLen+encrypted
hopList	Daftar CH yang harus dilalui pull request dalam urutan tertentu
typeFlags	Byte kombinasi message type (6-bit) + FLAG_ALARM_ACK + FLAG_GLD_EXT_POWER
pglTopology	Flow context Node-RED untuk menyimpan state jaringan CH/GW
NVS	Non-Volatile Storage ESP32 — penyimpanan mode, nulling profile, parent ID
TCXO	Temperature-Compensated Crystal Oscillator — sumber clock akurat SX1262

B. Daftar File Penting

File	Peran
firmware/platformio.ini	Definisi env build runtime
firmware/shared/include/ProtocolConstants.h	Semua konstanta protokol
firmware/shared/include/FirmwareVersion.h	Versi firmware semua komponen
firmware/config/GldConfig.h	Config GLD node
firmware/config/ChConfig.h	Config CH node
firmware/config/GwConfig.h	Config Gateway

File	Peran
firmware/config/ServerConfig.h	Config WiFi/MQTT server
firmware/config/LoraStarConfig.h	Config radio STAR
firmware/config/LoraMeshConfig.h	Config radio MESH
firmware/gld/src/GldUnifiedMain.cpp	Main runtime GLD
firmware/gld/src/GldNullingService.cpp	Nulling calibration
firmware/ch/src/ChStarMeshRuntimeMain.cpp	Main runtime CH
firmware/ch/src/ChRuntime.cpp	State machine CH
firmware/gateway/src/GatewayMqttMeshMain.cpp	Main runtime Gateway
server/nodered/functions/pertamina-gld-decode.js	Decode + AES-GCM server
server/nodered/apply-pertamina-gld-flow.js	Generator Node-RED flow
server/nodered/pertamina-gld-server.flow.json	Snapshot flow server

C. Daftar Env Aktif

Env	Board	Firmware	Tujuan
gld	4D-ESP32S3-Gen4	GLD 0.8.12	GLD runtime standar
gldw	ESP32-S3-WROOM-1U-N16R8	GLD 0.8.12	GLD runtime WROOM 16MB
ch1	4D-ESP32S3-Gen4	CH 0.7.1	CH ID 0x0064
ch2	4D-ESP32S3-Gen4	CH 0.7.1	CH ID 0x0065 (batt disabled)
ch3	4D-ESP32S3-Gen4	CH 0.7.1	CH ID 0x0066
gw	4D-ESP32S3-Gen4	GW 0.1.3	Gateway

D. Daftar MQTT Topics

Topic	Arah	Komponen
gld/gateway/uplink	GW → Server	Uplink frame MESH wrapped JSON
gld/gateway/topology	GW → Server	Topology CH_HELLO/Config JSON
gld/gateway/status	GW → Server	Status GW periodic
gld/gateway/cmd/#	Server → GW	Command wildcard
gld/gateway/cmd/pull	Server → GW	Pull request JSON
gld/gateway/cmd/node	Server → GW	Node command JSON
gld/server/decoded	Node-RED internal	Decoded normal GLD event
gld/server/alarm	Node-RED internal	Decoded alarm event
gld/server/topology	Node-RED internal	Topology event
gld/gateway/error	Node-RED internal	Error event
gas-leak-detector/F001/cmd	Server → GLD	Command dataset mode
gas-leak-detector/F001/dataset	Server → GLD	Dataset start/stop
gas-leak-detector/F001/dataset/data	GLD → Server	Data scan dataset

Topic	Arah	Komponen
gas-leak-detector/F001/dataset/status	GLD → Server	Status dataset
gas-leak-detector/F001/dataset/summary	GLD → Server	Summary sesi
gas-leak-detector/F001/cmd/ack	GLD → Server	Command ack

E. Daftar Message Types

Hex	Constant	Penggunaan
0x10	MSG_SENSOR_DATA	GLD uplink, alarm push, recovery, compact ACK
0x14	MSG_NODE_DOWNLINK	CH → GLD downlink
0x30	MSG_SERVER_PULL_REQUEST	Server pull request
0x31	MSG_CLUSTER_DATA_RESPONSE	CH pull response
0x32	MSG_SERVER_NODE_COMMAND	Server command ke GLD via CH
0x33	MSG_CH_HELLO	CH topology hello
0x34	MSG_CH_CONFIG_REQUEST	CH parent discovery
0x35	MSG_CH_CONFIG_RESPONSE	CH/GW discovery response

F. Daftar Log Token Penting

Token	Komponen	Arti
[BOOT_IC_REPORT]	GLD	Tabel IC check boot
[BOOT_SENSOR_SAMPLES]	GLD	5 baris sampel sensor
MODE_READY	GLD	Mode siap
NULLING_SERVICE_START	GLD	Nulling dimulai
NULLING_CH_OK / NULLING_CH_FAIL	GLD	Per-channel nulling result
NULLING_RUNTIME_RESULT=PASS	GLD	Semua channel sukses
PARTIAL_RETRY	GLD	Ada gagal, retry
CH_STATE	CH	State machine transition
CH_PARENT_SELECT	CH	Parent terpilih
CH_STAR_RX	CH	Terima dari GLD
CH_ALARM_ACK_RECV	CH	ACK alarm diterima
CH_ALARM_ACK_TIMEOUT	CH	ACK alarm timeout
CH_HELLO_TX	CH	Kirim topology hello
CH_CACHE_SUMMARY	CH	Ringkasan cache
CH_MESH_TX_KIND	CH	Jenis frame MESH dikirim

Dokumen ini dibuat berdasarkan source code dan konfigurasi repo Pertamina GLD per tanggal 2026-06-29.

*Source priority: `firmware/platformio.ini` → `shared/include/*.h` → `config/*.h` → `gld/, ch/, gateway/` → `server/nodered/` → `docs/design/*/final_design.md*`